Brian Hepler, PhD

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PROFESSIONAL SUMMARY

Data scientist with expertise in machine learning, network analysis, and statistical modeling. Built topic modeling platform analyzing 120K+ documents and network analysis revealing community organization patterns. Ph.D. in Mathematics with strong Python and SQL skills, database optimization experience, and proven technical communication abilities.

TECHNICAL SKILLS

- **Languages:** Python, SQL (PostgreSQL, SQLite), LaTeX
- **ML & Data Libraries:** Scikit-learn, PyTorch, XGBoost, NumPy, SciPy, Pandas, Matplotlib, NetworkX
- **Tools & Platforms:** Jupyter, Git, Heroku, Shiny, ipywidgets

SELECTED PROJECTS

Mathematical Research Intelligence Platform (GitHub [link], mathresearchcompass.com)

- Developed research intelligence platform analyzing 121,391 mathematics papers through advanced NLP pipeline (BERTopic, UMAP, HDBSCAN) and Claude AI-enhanced labeling. System discovers and categorizes research themes across 1,938 distinct topics, solving the challenge of navigating complex technical literature.
- Implemented SQLite database with indexed queries and LRU caching to handle interactive filtering across 121,391 papers with subsecond response times.
- Deployed interactive web application (Shiny on Heroku) enabling exploration of research themes, topic relationships, and author collaboration patterns across the full dataset.

Strategic Collaboration Network Analysis (GitHub [link], ArXiv [link])

- Investigated how research communities self-organize by analyzing collaboration patterns across 121,391 papers spanning 1,938 topics. Network analysis revealed structural patterns based on field popularity that traditional bibliometric approaches miss.
- Found that popular fields exhibit distributed, modular collaboration structures while niche areas show hierarchical organization around key figures. This structural dichotomy quantifies how communities scale differently with implications for organizational design and market analysis.
- Submitted manuscript to peer-reviewed journal *Scientometrics* and published arXiv preprint communicating findings through rigorous statistical analysis and research-quality documentation.
- Engineered analytical pipeline following software engineering best practices: automated builds (Makefile), modular Python components, and checkpoint recovery. Architecture supports reliable execution of complex multi-stage analysis at scale.

PROFESSIONAL EXPERIENCE

Quantum Formalism — *Remote*
Mathematical Consultant
Nov. 2024 – Present

- Consulted with machine learning and quantum computing researchers to identify relevant mathematical frameworks and translate advanced analytical concepts into practical applications and computational implementations
- Engineered interactive computational tools and visualization systems in Python (using Scikit-learn, PyTorch, PennyLane, Matplotlib, and ipywidgets) demonstrating quantum and

classical ML algorithms for diverse technical audiences
- Designed structured technical programs connecting abstract mathematical concepts to concrete problems in quantum computing and machine learning, adapting communication approach for stakeholders with varying levels of mathematical sophistication

Institut de Mathématiques de Jussieu-Paris Rive Gauche — *Paris, France*
Postdoctoral Researcher
2023 – Sept. 2024
- Contributed to competitively-funded international research project (Fondation Sciences Mathématiques de Paris) bridging multiple analytical frameworks to address problems in algebraic analysis
- Collaborated with international research group across different mathematical specializations, requiring synthesis of diverse methodological approaches and navigation of complex team dynamics
- Presented technical findings at research seminars and conferences, adapting communication style for audiences with varying technical backgrounds

University of Wisconsin–Madison — *Madison, WI*
Van Vleck Visiting Assistant Professor
2019 – 2023
- Published novel analytical framework in top-tier mathematics journal (Publ. RIMS Kyoto Univ.) after multi-year research effort, demonstrating sustained problem-solving on complex technical challenges
- Secured competitive $5,000 AMS-Simons research grant by identifying gaps in existing analytical approaches, proposing rigorous methodology, articulating project value to expert reviewers
- Designed and delivered undergraduate and graduate mathematics curricula, translating advanced technical concepts into structured learning experiences for students with diverse preparation levels

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SELECTED PUBLICATIONS (Google Scholar [link])

- Hepler, Brian. “Modular versus Hierarchical: A Structural Signature of Topic Popularity in Mathematical Research”, *arxiv e-prints*. DOI: 10.48550/arXiv.2506.22946
- Hepler, Brian and Hohl, Andreas. “Moderate Growth and Rapid Decay Nearby Cycles via Enhanced Ind-Sheaves”, *Publ. RIMS Kyoto Univ.*, 61 (2025), no. 1, pp. 1-51. DOI: 10.4171/prims/61-1-1
- Hepler, Brian. “Deformation Formulas for Parameterizable Hypersurfaces”, *Annales de l’Institut Fourier*, Volume 74, No. 3, pp. 1153−1188. DOI: 10.5802/aif.3613

EDUCATION

Ph.D., Mathematics — Northeastern University, 2019
Specialization: Algebraic Topology, Computational Geometry
M.S., Mathematics — Northeastern University, 2014
B.A., Mathematics — Boston University, 2012
Cum Laude with Distinction in Mathematics